

DEMENTIA PREDICTION

Artificial Intelligence Applications in Biology Final Project
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Dementia

Background on Dementia

-Dementia refers to the collection of symptoms one may have, like the loss of the ability to remember, think, and make decisions, which affect daily life.

-The most common cause of dementia is Alzheimer's disease, and around 7 million americans live with Alzheimer's.¹

-Age, Family History, Race/Ethnicity, Poor Heart Health, and Traumatic Brain Injuries can increase one's risk for dementia.²

Project Goal

To create a few machine learning models that can accurately predict if a patient has dementia and determine which model works the best.

Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import os
from sklearn import preprocessing
import numpy as np
import seaborn as sns

!pip install scikit-learn

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import roc_curve, auc
from sklearn.tree import plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
from sklearn import svm
```

Loading the Data

- The dataset was found on kaggle.
- The dataset had 24 features(columns), some continuous data and others categorical data.

Data Preprocessing

Label Encoding vs. One-Hot Encoding

- The first problem that I came across with this project was figuring out whether I use one-hot encoding or label encoding.
- After a few unsuccessful attempts at trying one-hot encoding I realized the I needed to use label encoding.
- The label encoding code that I used converted the categorical data into continuous data and also handled any missing values (NaN) by assigning them to their own distinct integer label.

Correlation Heat Map

- To reduce the chances of overfitting, I used a heat map to see which features had the highest correlations. The 3 features that had the highest correlation were Prescription, Dosage in mg, and Cognitive Test Scores.
- At first, after removing null values and converting categorical data into numerical data, I ran the first model and found that there was overfitting. I researched why this might be. I looked at the dataset on an excel sheet and noticed that 'Prescription' and 'Dosage in mg' had a high correlation with each other and with Dementia. I then decided to use a correlation heat map to confirm whether or not the two features should be taken out.
- the heat map allowed me to easily see which features were in high correlation with each other.

Data Splitting

- In this code, my independent variable is X. Y is my dependent/target variable (Dementia)
- test_size=0.3: 30% of the data will be used for testing, and 70% will be used for training.
- random_state=201: Creates reproducibility with the data. Allows for the data to be split the same way each time the code is run.
- X_train and y_train: Will train the machine learning model.
- X_test and y_test: Will evaluate the performance of the trained model.

```
X = dementia_data2.drop('Dementia', axis = 1)
y = dementia_data2['Dementia']
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size = 0.3, random_state = 201)
```

Models

Supervised Learning

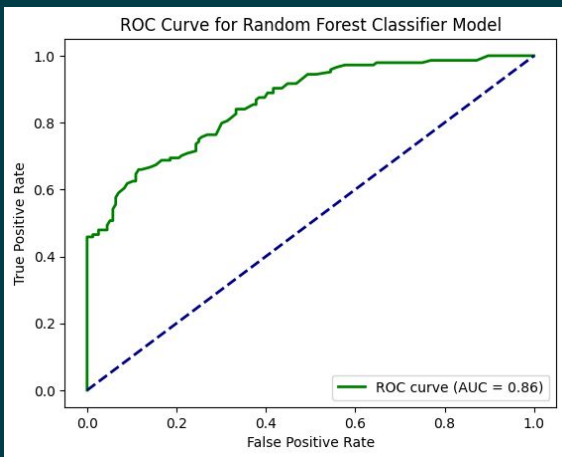
- Random Forest Classifier
- Logistic Regression
- Support Vector Machine

Results

ROC and AUC

Receiver Operating Characteristic Curve (ROC Curve) - a diagnostic plot for summarizing the behavior of a model by calculating the false positive rate and true positive rate for a set of predictions by the model under different thresholds.³

Area Under the Curve (AUC) - indicates the performance of the model and its ability to diagnose patients with and without the condition



Random Forest Classifier

AUC: 0.86

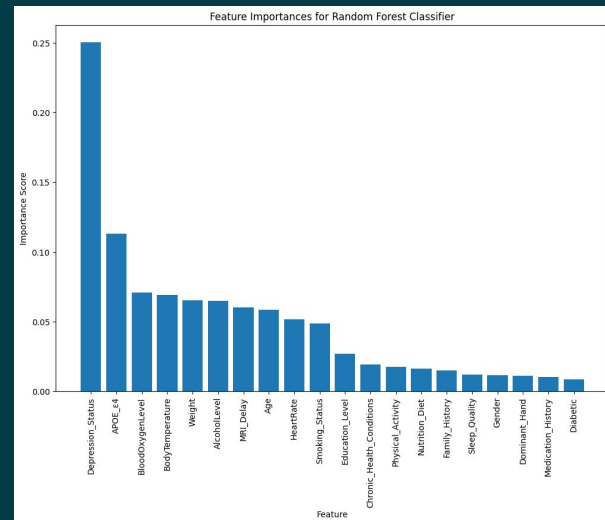
Feature Importance: The 'Depression Status' feature had a higher importance on the random forest classifier than any other feature, this may suggest that there is a strong association to depression and dementia. I would have to remove this feature so that every feature acts on the model equally.

Precision: 0.85

Recall: 0.65

F1: 0.73

Accuracy: 0.77



Results

Logistic Regression

AUC: 0.84

Precision: 0.82

Recall: 0.60

F1: 0.70

Accuracy: 0.75

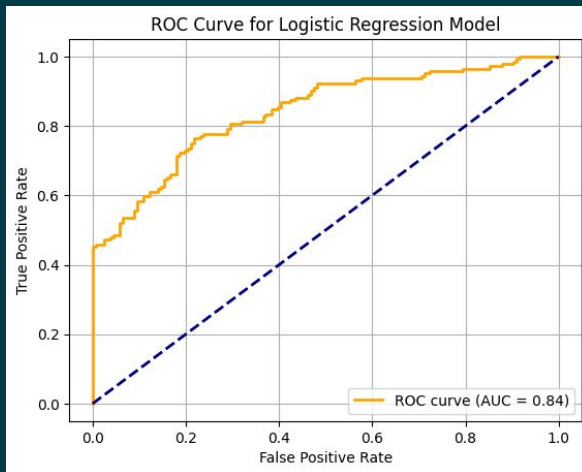
Support Vector Machine

Precision: 0.75

Recall: 0.62

F1: 0.68

Accuracy: 0.72



Conclusion

- Out of the three models, I believe that random forest classifier worked the best. It should be noted however that the features 'Depression_Status' and 'APOE_ε4' may have created a bias.
- In the future I would like to remove those two features, and play around with the dataset a bit to see the different results I would get

Citation

1. <https://www.alz.org/alzheimers-dementia/facts-figures>
2. <https://www.cdc.gov/aging/dementia/index.html>
3. <https://machinelearningmastery.com/tour-of-evaluation-metrics-for-imbalanced-classification/>
4. <https://www.kaggle.com/datasets/kaggler2412/dementia-patient-health-and-prescriptions-dataset>